AAA-B-92D
November 23, 1977
SUPERSEDING
Fed. Spec. AAA-B-92C
April 3, 1968

# FEDERAL SPECIFICATION

## BALANCES, ANALYTICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers analytical balances of 20 gram (g) and larger capacities with or without direct reading figures.

1.2 Classification. The balances covered by this specification shall be of the following classes and grades as specified (see 6.2).

Class 2 - Balances with quick weighing mechanism (see table I).
- Grade H - 200 g capacity, 0.2 maximum allowable sensitivity
- Grade I - 200 g capacity, 0.4 maximum allowable sensitivity

Class 3 - Balances with direct reading figures, and single load receiver (see table III).
- Grade L - 20 g minimum capacity
- Grade N - 80 g minimum capacity
- Grade O - 160 g minimum capacity

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-B-1122 - Balances, Scales, and Accessories, Packaging and Packing of.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specification and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

ASTM E 617 - Specification for Laboratory Weights and Precision Mass Standards

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Fire Protection Association (NFPA)

No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)



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National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The balances delivered under this specification shall be the manufacturer's commercial product except for any changes necessary to comply with specification requirements. All like items furnished on any one contract, including parts and subassemblies thereof, shall be interchangeable.

3.2 Materials. Materials not definitely specified shall be of the quality used in the manufacturer's commercial product.

3.3 Design and construction. All balances shall be provided with the following features conforming to the manufacturer's current commercial practice or as otherwise specified herein:

a. Leveling indicator to indicate the level condition
b. Arresting devices for beam or weighing pans or both as applicable
c. Equilibrium adjuster to indicate the equilibrium position
d. Damping devices
e. Corrosion-resistant and non-magnetic beams, beam riders, removeable plate type load receivers and hangers
f. Case (see 3.3.1 and 3.3.2) constructed so as to provide isolation from drafts and thermal disturbance with access doors or removeable panels provided for operation and cleaning. The floor shall be either black polished glass or non-magnetic corrosion-resistant material
g. Dust cover, when required, constructed of clear material designed to be removeable and to completely cover the case except for the bottom (see 6.2)



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h. Electrical operation, when required, shall be 115 volt, 60 Hz AC with wiring installation conforming to NFPA Standard No. 70 (see 6.2)
i. Knife edge or torsional balance construction
j. Balance weights conforming to ASTM E 617
k. Any items necessary for the proper operation of the balances

3.3.1 Class 2. The class 2 balances shall be quick-weighing chain type conforming to the requirements of table I and as specified herein. The balance shall have an equal arm beam mounted on a column with plate type load receivers suspended from each end of the beam. The beam shall be notched and graduated from left to right from 0 to 1.0 gram in 0.1 gram divisions. The beam rider shall be of the vertical lift or pusher/roller type capable of maintaining rapid and accurate alignment with no evidence of rider dislodgement when moved in either direction using minimum effort. The controls for the balances shall be on the outside of the case in the bottom of which shall be a front opening drawer (see 6.2).

3.3.1.1 Chain. The chain balance shall have a chain with one end attached to the beam and the other attached to an indicating scale with vernier or digital readout. The drain shall be operated from outside the case. The indicating mechanism shall be graduated from 0 to chain capacity in accordance with table I.

TABLE I. Class 2 analytical balances

| Grade | Weighing capacity (grams) | Maximum allowable sensitivity, requirement (milligrams per division) | Maximum beam arm ratio accuracy (parts per million) | Minimum precision (milligram, see 3.4.3) | Chain weight range (milligram) |
|---|---|---|---|---|---|
| H | 200 | 0.2 | 5 | 0.1 | .005 to 1.0 |
| I | 200 | 0.4 | 10 | 0.2 | 0 to 100.0 |

3.3.1.2 Dimensions. Specific dimensions required for the class 2, grades H and I balances shall be as specified in table II.



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TABLE II. Dimensions

| Grade | Load receiver width (minimum) | Hanger width (minimum) | Hanger length (minimum) | Case (maximum) |
|---|---|---|---|---|
| H | 2-1/2 inches (61.9 mm) | - | 6-1/2 inches (165 mm) | - |
| I | 2-7/10 inches (63.5 mm) | 4-1/8 inches (105 mm) | 8 inches (203 mm) | 20-1/2 inches high x 19-1/2 inches wide x 10-1/2 inches deep (521 mm x 495 mm x 267 mm) |

3.3.2 Class 3. The class 3 balances shall conform to table III and as specified herein. The balances shall be front and direct reading, single plate receiver, constant load and complete with built-in weights. Both the built-in weights and the hanging plate shall be on the same side of the beam and shall operate with a constant load on the beam. The balance shall operate by subtracting built-in weights from the beam. The total of the weights shall be included in the direct reading figures. The weights shall be controlled by knobs. If the balance is of knife edge construction, a beam arrest and plate arrest shall be provided. A light to illuminate the direct reading figures may be furnished. If applicable, a single multi-purpose control or individual controls shall operate the beam and plate arrests and direct reading figure lights. All controls shall operate outside the case and shall be marked with use or weight value. The balances shall have zero adjustment capability. The balance shall be enclosed in a case with a removeable top.

3.3.2.1 Built-in weights. The built-in weights for the class 3 balances shall conform to the applicable requirements specified in ASTM E 617.

TABLE III. Class 3 analytical balances

| Grade | Weighing capacity minimum (grams) | Maximum sensitivity (milligram) | Minimum precision (milligram) | Readability (milligram) |
|---|---|---|---|---|
| L | 20 | 0.005 | 0.001 | 0.001 |
| N | 80 | 0.01 | 0.01 | 0.01 |
| O | 160 | 0.1 | 0.03 | 0.05 |



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3.4 Performance.

3.4.1 Knife edge construction. When knife edge construction is furnished there shall be no sliding motion or vibration between knife edges and bearings when beam arrest is tested as specified in 4.3.1.

3.4.2 Capacity. The capacities shall be as specified in tables I and III. The balance, when tested as specified in 4.3.2 with capacity loads, shall show no bends, deformations, misalignments of bearings or knife-edges and shall return to the precision of the balance when the load is removed with the same value of precision as specified in 4.3.

3.4.3 Precision. The precision, which is the degree of agreement of repeated weighing of the same quantity, specified in tables I and III shall be tested as specified in 4.3.3.

3.4.4 Readability. The smallest weight change that can be read directly from the balance indicators specified in table III shall be tested as specified in 4.3.4.

3.4.5 Sensitivity requirement. The sensitivity requirement specified in table I shall be tested as specified in 4.3.5.

3.4.6 Beam arm ratio. The beam arm ratio, in parts per million, specified in table I shall be tested as specified in 4.3.6.

3.4.7 Weighing chain. The class 2 balances, furnished with a chain type mechanism, shall be tested as specified in 4.3.7 to determine that accuracy of readings are within $\pm$ 0.2 mg when obtained by use of the chain mechanism.

3.4.8 Rider notches. Balances having beams with notches shall be tested as specified in 4.3.7 to determine that indicated values are within $\pm$ 0.2 mg.

3.5 Finish. The balance shall have all surfaces finished in accordance with the manufacturer's commercial practice.

3.6 Marking for identification. Each balance shall be permanently and legibly marked with the manufacturer's name or readily identifiable trademark.

3.7 Workmanship.

3.7.1 Metal and plastic components. All metal or plastic components shall be free of sharp edges, cracks, splinters, kinks, or sharp bends. There shall be no fractures, splits, punctures, or dents.



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3.7.2 Glass. Glass shall fit securely in frames and shall be free of cracks, splinters, chips, or discoloring.

3.8 Contract data requirements. Contract data requirements shall be as specified (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.1.1 Corrosion-resistant and non-magnetic material. The beam and beam riders and load receivers and hangers shall be inspected to determine compliance with the requirements specified in 3.3, as applicable. The case shall be inspected for drafts and thermal disturbances. The lot shall be all similar items offered for inspection at one time. The sample unit shall be one item. Nonconformance of any item shall be cause for rejection of the whole lot of that item.

4.2.2 End item inspection. A lot shall consist of all balances of the same class and grade offered for inspection at one time. The sample unit for these inspections shall be one completely fabricated balance.

4.2.2.1 Visual examination. Examination of the end item shall be in accordance with the classification of defects in table IV. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.



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TABLE IV. Classification of defects

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Design | Any characteristics not in accordance with the specified requirements | X | |
| Construction and workmanship (applicable to all components and assemblies) | Part missing or not specified type | X | |
| | Any component not readily accessible for servicing where required | X | |
| | Component not properly assembled or secured | X | |
| | Unit perceptibly out of square alignment | X | |
| | Fractured, malformed, or otherwise impaired | X | |
| | Sharp burr, sliver, or splinter | | X |
| Knife-edges and bearings. (when applicable) | Not ground and polished | X | |
| Beam arrest | Not operated from outside case | X | |
| Torsion bands | Not in accordance with manufacturer's current commercial practice | X | |
| Equilibrium indicator | Equilibrium indicator missing | X | |
| Graduated scale | Missing or not as specified (when applicable) | X | |
| Glass floor (when applicable) | Cracks, splinters, or loose chips | X | |
| | Not black color | | X |
| Controls and keys | Not on the outside of the case | X | |
| | Not marked with use or weight value they represent | | X |
| Direct reading figures (when applicable) | Not on front of case | | X |



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TABLE IV. Classification of defects (cont'd)

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Cover | Not as specified | X | |
| | Not removeable or does not completely enclose the exterior of the case | | X |
| Case | Not as specified | X | |
| | Doors hard to open | | X |
| Electrical | Wiring not secured | | X |
| | Wires cut or abraded, not properly joined | | X |
| Marking (identification) | Missing, incomplete, not legible, not specified type or size, not in proper location, not affixed in the prescribed manner | | X |

4.2.2.2 Dimensional examination. Inspection shall be made of the end item for compliance with dimensions specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.2.2.3 Testing of the end item. The end item shall be tested as specified in 4.3.1 through 4.3.9, as applicable. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units, except for 4.4.5 sensitivity requirement which shall be tested on each class 2 balance (see 6.2).

4.2.3 Inspection of preparation for delivery. Preparation, packaging, packing and marking shall be inspected in accordance with the quality assurance provisions in PPP-B-1122.

4.3 Tests. The balance or balances for the tests specified hereinafter shall be placed in operating position in a location free from drafts and vibrations and at a constant temperature and humidity.

4.3.1 Beam arrest. Weights equal to the capacity of the balance shall be placed on pans and the beam arrest, when applicable, operated to determine compliance with 3.4.1.



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4.3.2 Capacity. Weights equal to the capacity of the balance shall be applied to each load receiver and then removed and the balance observed to determine compliance with the requirements specified in 3.4.2.

4.3.3 Precision. The precision shall be calculated at the full capacity load of the balance to determine compliance with the requirements specified in 3.4.3. A load equal to full capacity of the balance shall be weighed and the value recorded. This shall be repeated 10 times. The precision (P) shall then be calculated for each load using the following formula:

$$P=\sqrt{\frac{\text{SUM of } (d)^2}{f}}$$

(The letter (d) in the formula is the average of all the weighings minus an individual weighing and (f) is the number of weighings minus one).

4.3.4 Readability. Readability of the balances shall be tested at zero load, 1/2 capacity, and full capacity load of the balance. Weights equal to these loads shall be placed on the plate and the balance observed to determine compliance with the requirements specified in 3.4.4.

4.3.5 Sensitivity requirement. The sensitivity requirement for the class 2 balances shall be tested at loads equal to zero and 1/2 capacity of the balance. Weights equal to these loads shall be applied on the load receivers. A two milligram weight or a weight equal to four times the maximum sensitivity requirement allowable, whichever is larger, shall be applied to one load receiver and the sensitivity requirement calculated to determine compliance with the requirements specified in 3.4.5. Sampling shall be as specified (see 6.2).

4.3.6 Beam arm ratio. The beam arm ratio shall be determined by observing the difference in scale divisions between the zero point and the mean rest point when equal loads are interchanged on the load receivers. The ratio of this difference, converted to milligrams using the sensitivity requirement of the balance at the given load, to the load used in milligrams shall be the beam arm ratio expressed in parts per million. The beam arm ratio shall be obtained using zero load and full capacity load of the balance to determine compliance with the requirements specified in 3.4.6.

4.3.7 Weighing chain test. The balances with chains shall be tested with zero load, 50 mg load and 100 mg load using standard weights. After each reading, the balance shall be examined for compliance with the requirements specified in 3.4.7.



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4.3.8 Rider notches. The balances shall be tested with standard weights at not less than four different points on the notched beam. After reading, the balance shall be examined for compliance with the requirements specified in 3.4.8.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. The balances shall be preserved and packaged as specified for level A in PPP-B-1122.

5.1.2 Commercial. Each balance shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling and storage.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The balances preserved and packaged as specified in 5.1.1 shall be packed as specified for level A in PPP-B-1122.

5.2.2 Level B (Military requirements). Balances, preserved and packaged as specified in 5.1, shall be packed as specified for level B in PPP-B-1122.

5.2.3 Commercial. The packaged balances shall be packed in shipping containers to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Unitization (Civil agencies). When shipments to Government depots are full car or truck load, the shipping containers shall be unitized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds (1116 kg) in weight, 63 inches (160 cm) in height, 56 inches (142 cm) in length, and 45 inches (114 cm) in width.

5.4 Marking. Marking shall be in accordance with 5.4.1 or 5.4.2 as specified (see 6.2).



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5.4.1 Civil agencies. Packages, shipping containers and unitized loads (when applicable) shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military agencies. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The analytical balances covered by this specification are intended primarily to be used to weigh within the capacities specified.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class and grade of balance required (see 1.2, tables I and III).
(c) When drawer is required in case (see 3.3.1).
(d) When dust cover is required (see 3.3 g.).
(e) When electrical characteristics are required (see 3.3 h.).
(f) When contract data requirements are required (see 3.8 and 6.2.1).
(g) When sensitivity testing is required for each class 2 balance (see 4.2.2.3).
(h) Selection of the applicable levels of preservation and packaging and packing (see 5.1 and 5.2).
(i) Whether Civil or Military marking is required (see 5.4).

6.2.1 Contract data requirement. Any requirement for contract data for the balances covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

6.3 Supersession data. The balances covered by this revision AAA-B-92D have been revised from those covered by the previous revision as follows:

| AAA-B-92D | AAA-B-92C |
|---|---|
| Deleted | Class 1 balances (all grades) |
| Deleted | Class 2, grades HH, HHH, HHHH and HHHHH; II and III; and K, KK, and KKK |
| Deleted | Class 3, grade M |

The deleted balances are no longer required.



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Army - GL
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Review activities:

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User activity:

Navy - MS

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA - FSS
HEW - NIH
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